

Data-Driven Cloth for Interactive Applications

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1. Problem statement

Physics-based cloth simulation is accurate but far too slow for interactive uses such as games, e-commerce try-on or VR. Our assigned problem is to reproduce the data-driven approach of Xue & Wu (*Data-Driven Clothing for Interactive Applications*, Stanford CS229, 2021): learn a fast model that takes a character’s pose and predicts how the clothing deforms. Concretely, the input is the pose (14 joint rotations as quaternions, $14 \times 4 = 56$ numbers) and the output is the offset of every cloth vertex between the cheap skinned cloth and the simulated cloth (3 numbers per vertex). We compare a neural network that predicts PCA coefficients of these offsets against a linear-regression baseline and a network that predicts the offsets directly.

2. Dataset

Original data. The paper uses 10,000 random poses of a coat (2,170 vertices, 6,510 outputs), skinned in Blender and simulated with PhysBAM; 187 faulty simulations are removed with a distance threshold, leaving 9,813 examples split 6,000 / 2,000 / 1,813. This dataset and the simulator are not public, so we could not use it.

Our data (synthetic stand-in, same format). We generate 10,000 poses by drawing a random axis and angle for each of the 14 joints (per-joint limits of 30–90°) and converting to unit quaternions (56 inputs). The “simulated” cloth is a 20×25 vertex panel (1,500 outputs); its ground-truth offsets are a fixed *nonlinear* function of the pose (two tanh layers feeding 64 smooth low-frequency spatial modes, scaled to about 1.5 cm), plus Gaussian noise ($\sigma = 0.05$ cm). About 1% of examples are deliberately blown up ($\times 6$ – 12) to mimic faulty simulations. Like the paper, we remove examples whose largest vertex displacement exceeds a threshold (median + 5 robust standard deviations), then split 6,000 train / 2,000 validation / the remaining ≈ 1.9 k test. PCA is fitted on the training set only. **All results are therefore on synthetic data;** they test the pipeline, not real cloth physics.

3. Approach

- **PCA on offsets.** Offsets are centred and projected onto the top- k principal components (computed by SVD of the training offsets): $x \approx \sum_1 c_i u_i + \mu$. This shrinks the output dimension ($1,500 \rightarrow k$) and acts as a regulariser.
- **Network.** A fully-connected network $56 \rightarrow 128 \rightarrow 256$ (ReLU) $\rightarrow k$ PCA coefficients, followed by a fixed (non-trainable) linear “PC layer” that reconstructs the 1,500 vertex offsets, so the loss is the MSE on offsets: $L = (1/n) \sum (x_i - p_i)^2$, where p_i is the predicted offset. Adam, learning rate $1e-4$, batch size 32.
- **Baseline.** Linear regression (a network with no hidden layers, $56 \rightarrow 1,500$), solved in closed form by least squares.
- **Ablation.** Same network predicting offsets directly (no PCA) and with $k = 256, 128$ and 64 components.

4. Implementation overview

The code is in PyTorch/NumPy/Matplotlib in a private GitHub repository (see its README for setup). `src/data.py` generates the data, filters outliers and splits it; `src/pca.py` fits PCA on training offsets; `src/model.py` defines the MLP with the fixed PCA layer and the closed-form baseline; `src/train.py` trains one model and keeps the checkpoint with the best validation loss; `src/evaluate.py` computes test metrics, timings and plots; `demo.py` is a live viewer that shows ground-truth and predicted cloth for a chosen test pose. All MLPs were trained for 1,000 epochs (as in the paper) with seed 0.

5. Results

Metrics are on the held-out test set with the best-validation checkpoint (single run). Test MSE is the mean squared error over all vertex coordinates; vertex error is the mean Euclidean distance per vertex.

Model	Output dim	Trainable params	Test MSE (cm ²)	Mean vertex err. (cm)	Train time
Predict train mean	–	0	1.831	–	–
Linear regression	1,500	85,500	0.392	0.991	closed form
MLP, direct offsets	1,500	425,820	0.251	0.789	537 s
MLP + PCA, $k = 256$	256	106,112	0.253	0.794	520 s
MLP + PCA, $k = 128$	128	73,216	0.254	0.795	459 s
MLP + PCA, $k = 64$	64	56,768	0.250	0.790	532 s

Table 1: Ablation on the test set. Predicting the training mean gives MSE 1.831.

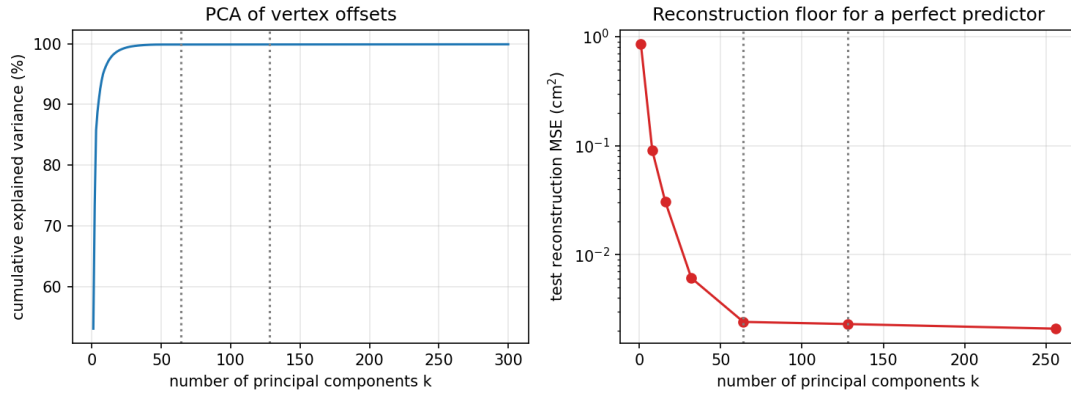


Figure 1: Cumulative explained variance of the offsets (left) and test reconstruction error of a perfect predictor (right) vs number of principal components k ; dotted lines mark $k = 64$ and 128 .

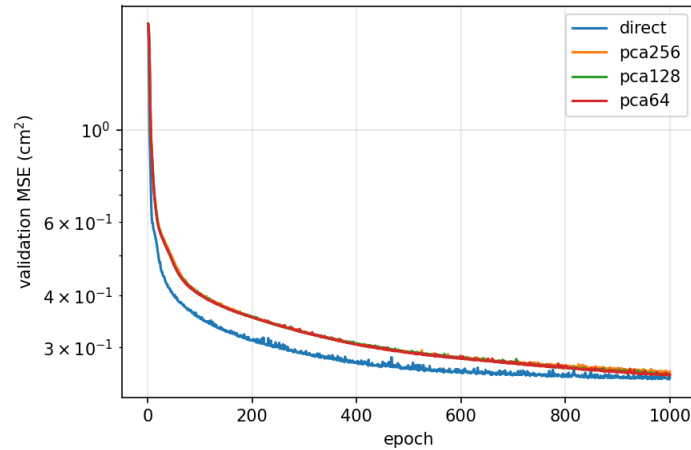


Figure 2: Validation MSE during training (log scale). All curves have nearly flattened by epoch 1,000.

6. Conclusions

- The nonlinear MLPs clearly beat linear regression: test MSE falls by 35–36% ($0.392 \rightarrow 0.250\text{--}0.254$) and the mean vertex error from 0.99 to about 0.79 cm, so pose-to-cloth deformation is not captured well by a linear model.
- **PCA costs no measurable accuracy but shrinks the network.** All four MLPs are within 1.5% of each other (MSE 0.250–0.254), which is within the noise of a single run, so we cannot rank direct prediction against PCA or the three values of k . With $k = 64$ the network has 56.8k trainable parameters versus 425.8k for direct prediction (7.5× fewer). Already 64 components keep $\approx 99.8\%$ of the variance and their reconstruction error (≈ 0.002) is far below the model error, so extra components add nothing here.
- **Comparison with the paper.** Xue & Wu report a linear baseline about $2.6\times$ worse than their best network, with $k = 128$ best (more components overfit, fewer underfit) and PCA about 6% better than direct prediction. We reproduce the main trends (MLPs clearly beat linear regression, $1.6\times$ in our case; PCA does not hurt accuracy and shrinks the model) but not a preferred k or a clear PCA advantage. A likely reason (not tested) is that our synthetic offsets are smooth and almost exactly low-rank, so there is less to overfit than with real simulated cloth. Absolute errors are not comparable with the paper’s because the data and units differ.
- All models take roughly 0.06–0.2 ms to predict one pose on CPU (timings varied between runs, so we do not rank them), far below a 16 ms frame budget, which supports the use of such networks for real-time cloth.
- **Limitations / future work:** the data is synthetic and every result is from a single random seed; several seeds would show whether the small differences between k values are real, and real simulated cloth (e.g. a mass-spring simulation) would be needed to test whether the paper’s PCA advantage appears.

Reference: K. Xue and J. Wu, “Data-Driven Clothing for Interactive Applications”, Stanford CS229 project report, Spring 2021 (cs229.stanford.edu/proj2021spr/report2/81970649.pdf).